Beyond the Single Split: A Reproducible Benchmark of Bengali Book Review Sentiment Classification with Explanations and Label-Noise Analysis

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***Abstract*—Reported accuracies for Bengali book-review sentiment are hard to compare: they often come from a single split of a small corpus, against weak baselines, with labels derived from star ratings. We re-examine the problem with repeated stratified cross-validation on a 2,009-review corpus (BBR-2009) rebuilt from the Mendeley and BanglaBook sources. Recurrent models trained from scratch reach at most 81.1% accuracy, and one model's accuracy swings by 13 points between folds, so a single split can tell very different stories. Frozen pretrained embeddings lift the same models to 87.6%. Fine-tuned BanglaBERT reaches $93.8 \pm 1.1\%$, ahead of XLM-R (91.3%), multilingual BERT (86.8%) and a TF-IDF SVM (90.2%), while a larger model and an ensemble add nothing statistically reliable. On the Mendeley corpus BanglaBERT reaches $96.7 \pm 1.2\%$, yet a plain TF-IDF Naive Bayes already reaches 94.2%, far above the 84% reported earlier on a larger corpus. SHAP explanations pass a deletion test. Of the model's errors, 62% are confident disagreements concentrated in rating-derived labels (8.6% of BanglaBook reviews against 1.3% of Mendeley reviews), and a blind audit by one LLM annotator suggests that many are label errors. On the official BanglaBook benchmark we reach 0.927 weighted F1, 0.6 points below the published value, so we make no state-of-the-art claim.**

***Index Terms*—Bengali sentiment analysis, book reviews, BanglaBERT, cross-validation, label noise, SHAP**

I. Introduction

Readers increasingly post opinions about Bengali books online, and sentiment classification can make these reviews useful. The text is difficult, though: spelling varies, informal writing and emoji are common, and annotated data is scarce.

A second problem is less visible. Many published results on Bengali book reviews come from a single split of a small corpus. A test set of 400 reviews has a 95% confidence interval of roughly ± 2 points at 95% accuracy, so ranking models by one split says little. Our own cross-validation shows how large the effect is: the best recurrent model scores between 72.6% and 85.8% on different folds, so the reported figure depends heavily on which reviews are held out.

This paper asks a simple question: *what do the numbers look like when the evaluation is done carefully?* Our contributions are:

- A cross-validated benchmark of twelve recurrent models, nineteen classical classifiers and four pretrained transformers, run with several seeds and with significance tests and calibration (Sections III–IV).
- Evidence that single-split figures are unstable (one model's fold accuracies span 13 points), that a simple TF-IDF Naive Bayes already reaches 94.2% on the public Mendeley reviews, far above the 84% reported earlier on a larger corpus, and that Bengali-specific pretraining matters more than model size or recurrence.
- A label-noise analysis showing that confident model errors concentrate in rating-derived labels, supported by a blind re-annotation audit.
- An explainability study with a deletion-based faithfulness check, plus an honest comparison with the official BanglaBook and Rokomari benchmarks, where we make no state-of-the-art claim.

We also release the corpus construction, fold assignments and a datasheet.

II. Related Work

Early sentiment classifiers paired Naive Bayes, logistic regression or SVMs with TF-IDF features [1]. Recurrent networks [2], gated cells [3], [4], bidirectional reading [5] and attention [6], [7] then became standard, before the Transformer [8] and BERT-style pretraining [9] changed the field again. BanglaBERT [10] is an ELECTRA model pretrained on a large Bengali corpus; XLM-R [11] is a multilingual alternative.

For Bengali, Islam *et al.* applied multilingual BERT [12], Hasan *et al.* compared classical and neural models [13], and Bhowmick and Jana studied transformers [14]. On book reviews, Hossain *et al.* reported 84% with Naive Bayes on their own 2,000-review corpus [15], Sarker *et al.* reached 88.95% weighted F1 with XLM-R on Rokomari reviews [16], and BanglaBook [17] offers 158,065 rating-labeled reviews on which BanglaBERT-large scores about 0.93 weighted F1. To our knowledge, no study evaluates Bengali book-review classifiers with repeated cross-validation and significance tests, quantifies the effect of rating-derived labels, or checks whether token attributions are faithful on this task.

III. Data and Experimental Setup

A. Data

BBR-2009 contains 2,009 unique reviews (980 positive, 1,029 negative). It combines 1,414 reviews from the Mendeley Bengali Book Reviews corpus (manually labeled, CC BY 4.0) with 595 reviews from BanglaBook [17] (562 negative, 33 positive; star ratings 1–2 mapped to negative and 5 to positive; CC BY-NC-SA 4.0). It is a reconstruction from these public sources. The BanglaBook part is almost entirely negative, so

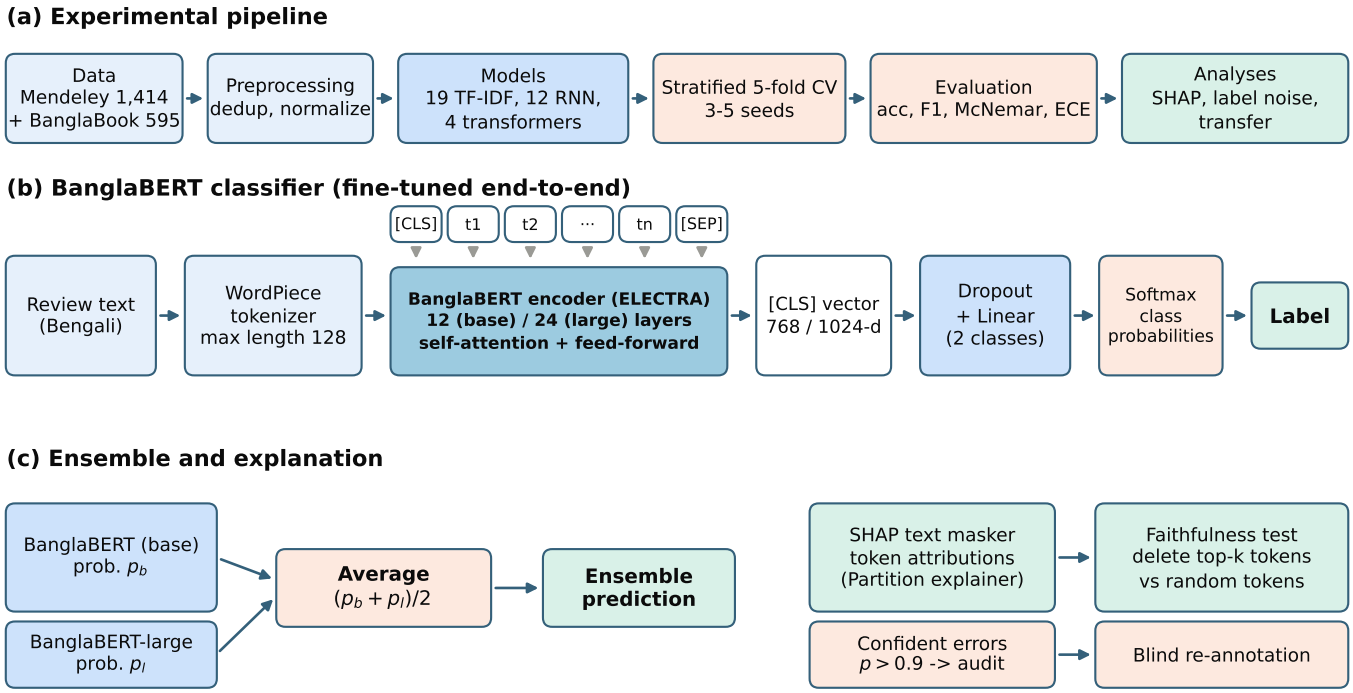


Fig. 1. Overview. (a) Experimental pipeline. (b) The BanglaBERT classifier: the review is tokenized, encoded by the ELECTRA encoder, and the [CLS] vector feeds a dropout and linear layer. (c) Base and large probabilities are averaged, and token attributions and confident errors are analyzed afterwards.

source and label are partly confounded, and we report results by source where it matters. We also use the Mendeley corpus alone (1,442 unique reviews, 970 positive and 472 negative), the Rokomari corpus [16] (5,636 positive and negative reviews: 3,051 and 2,585) and the official three-class BanglaBook split (110,645 / 15,806 / 31,614 reviews). A cleaned Rokomari subset (5,277 reviews, none shared with BBR-2009) serves as a zero-shot external test.

B. Models

Fig. 1 summarizes the pipeline and the BanglaBERT classifier. **Recurrent models:** RNN, GRU and LSTM, each unidirectional or bidirectional, with or without additive attention (twelve in total), using a 15,000-word vocabulary, 128-d trainable embeddings, sequences of 150 tokens, Adam (learning rate 10^{-3}), dropout 0.3, gradient clipping at 1.0 and early stopping on 10% of each training fold. **Classical models:** nineteen classifiers on word (1–2-gram) and character (2–5-gram) TF-IDF features, from logistic regression and SVMs to boosted trees and an MLP. **Transformers:** BanglaBERT (base) and BanglaBERT-large [10], XLM-R base and multilingual BERT, each fine-tuned end-to-end with AdamW, linear warm-up and decay, maximum length 128 and balanced class weights (four epochs; BanglaBERT 3×10^{-5} with batch 16, BanglaBERT-large 2×10^{-5} with batch 8 in cross-validation). XLM-R and mBERT were tuned over four learning rates and two epoch budgets on a tuning split taken from the training part of one fold, so evaluation folds were never touched. The ensemble averages the probabilities of the base and large BanglaBERT models.

C. Evaluation protocol

Every model uses stratified five-fold cross-validation (about 1,607 training and 402 test reviews per fold): five seeds for the recurrent models and three for the transformers and for the classical models on BBR-2009. We report accuracy and macro-F1 as mean $\pm$ standard deviation over runs, paired Wilcoxon tests over runs, McNemar's test [18] on pooled out-of-fold predictions, and the expected calibration error (ECE, ten bins) [19]. For the official BanglaBook split, every choice (model, ensemble members, class offsets) was made on the validation set and the test set was scored once. Explanations use SHAP [20].

IV. Results

A. Pretraining matters more than recurrence

Fig. 2 gives the overall picture on BBR-2009. The best of the twelve recurrent models, a BiLSTM with attention, reaches $81.1 \pm 3.8\%$, and the plain RNN only $73.6 \pm 6.2\%$. The spread between folds is large (3.6 to 6.7 points), which is why one split can mislead: the same BiLSTM with attention scores anywhere between 72.6% and 85.8% on different folds. A linear SVM on TF-IDF features ($90.2 \pm 1.5\%$) beats every recurrent model, so the extra machinery of recurrence and attention does not pay off on a corpus of this size.

The recurrent models, however, start from random embeddings, which is a handicap on 1,600 training reviews. To see how much it matters, we gave three of them (a BiLSTM, a BiLSTM with attention and a BiGRU with attention) the frozen input embeddings of BanglaBERT, with everything else

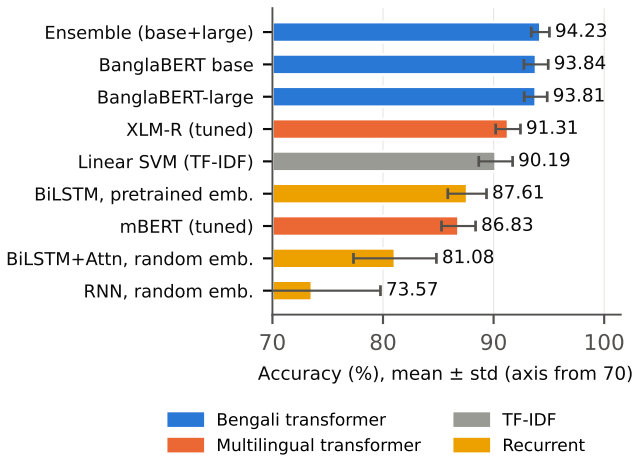


Fig. 2. Accuracy on BBR-2009 (three seeds × five folds; recurrent models five seeds). Bars show the mean and whiskers one standard deviation.

unchanged. Accuracy rises from 81.1% to $87.6 \pm 1.8\%$ for the BiLSTM (BiGRU with attention $87.4 \pm 2.1\%$, BiLSTM with attention $87.0 \pm 2.1\%$), and attention again adds nothing. Pretrained embeddings thus explain about six points of the gap to BanglaBERT; the remaining six come from its contextual encoder, and a TF-IDF SVM still sits between the two.

Pretraining does. Fine-tuned BanglaBERT reaches $93.8 \pm 1.1\%$ (macro-F1 93.8%) with a much smaller spread. Pooled over the folds of seed 0 it labels 95 negative reviews positive against 35 positive reviews negative, and the large model behaves the same way (81 against 34); negative reviews are the harder class.

One caution is needed. Because source and label are confounded, a classifier that knows only the source (positive for Mendeley, negative for BanglaBook) already scores 75.1%, so the recurrent models are only about six points above this shortcut. Inside the BanglaBook part, always answering "negative" gives 94.5%, more than BanglaBERT's 87.9% there, while on the Mendeley part BanglaBERT reaches 97.2% against 67.0% for the majority class. The overall figure therefore reflects mostly the Mendeley reviews, and we read the BBR-2009 results with that in mind.

B. Choosing among pretrained models

Bigger is not better here. BanglaBERT-large ($93.8 \pm 1.0\%$) is indistinguishable from the base model (Wilcoxon $p = 0.89$), and the base+large ensemble ($94.2 \pm 0.8\%$) is not significantly better either ($p = 0.16$; McNemar's test finds a difference in only one of the three seeds). Language-specific pretraining matters more than size: XLM-R reaches $91.3 \pm 1.1\%$ and multilingual BERT $86.8 \pm 1.5\%$, in that order in every seed, and tuning moved them by at most 0.3 points. McNemar's test confirms that the ensemble beats the best TF-IDF model in every seed ($p < 10^{-8}$). Calibration is reasonable (ECE 0.047, 0.057 and 0.041 for the base, large and ensemble models).

TABLE I
BANGLABERT ABLATION ON BBR-2009 (3 SEEDS × 5 FOLDS).

Variant	Accuracy (%)	Δ vs. default
Default (lr 3×10^{-5}, 4 epochs)	93.84 ± 1.10	–
Learning rate 5×10^{-5}	93.86 ± 1.31	$+0.02$
No class weights	93.83 ± 1.27	-0.01
Maximum length 64	93.26 ± 1.42	-0.58
Two epochs	93.13 ± 1.21	-0.71

Table I shows the result is not fragile: dropping class weights or raising the learning rate changes nothing, and shorter inputs or fewer epochs cost 0.6 to 0.7 points, within one standard deviation.

C. Other datasets and published results

Fig. 3(a) and Table II set our numbers beside published ones. On the Mendeley corpus BanglaBERT reaches $96.7 \pm 1.2\%$, but Multinomial Naive Bayes already reaches $94.2 \pm 1.7\%$ under our protocol, while the 84% reported earlier [15] comes from the authors' own 2,000-review corpus, of which only 1,443 reviews are public, so the comparison is indicative only. On Rokomari, BanglaBERT reaches $90.6 \pm 0.7\%$ accuracy in cross-validation. The authors of [16] report 88.95% weighted F1 on a split we cannot reproduce, because their shuffle is unseeded. Following their protocol (positive and negative reviews of at most 300 words, stratified 80/10/10) with three random shuffles, BanglaBERT obtains $89.97 \pm 0.66\%$ weighted F1. That is a point higher, but with a different split and about 500 test reviews it sits well inside the shuffle-to-shuffle variation, so we do not read it as an improvement.

The official BanglaBook split (Fig. 3(b)) is where we fall short. Our first runs reached 0.892 and 0.919 weighted F1, and one later run collapsed to the majority class (0.846); we adjusted those runs after seeing test scores and show them only for transparency. We then fixed the protocol and chose everything on the validation set. Six runs (five BanglaBERT-large runs with different seeds, learning rates and class weights, and one BanglaBERT base run) each scored between 0.923 and 0.926 on the test set. Choosing the ensemble members and class offsets on the validation set picked three of the large runs, and their average scored **0.927** (accuracy 93.6%, macro-F1 0.681). That is 0.6 points below the 0.933 reported in [17], and we make no claim to match it. Per-class F1 is 0.72, 0.36 and 0.97 (negative, neutral, positive) against 0.72, 0.40 and 0.97 in [17]; the gap lies in the neutral class.

D. What the model looks at

We computed SHAP values for all 402 held-out reviews of an 80/20 split. Fig. 4 shows the most influential tokens: words meaning *bad*, *annoying* and *worthless* push toward negative, and *best*, *excellent* and *great* toward positive. A few function words and word pieces also appear, which hints at some dataset-specific cues.

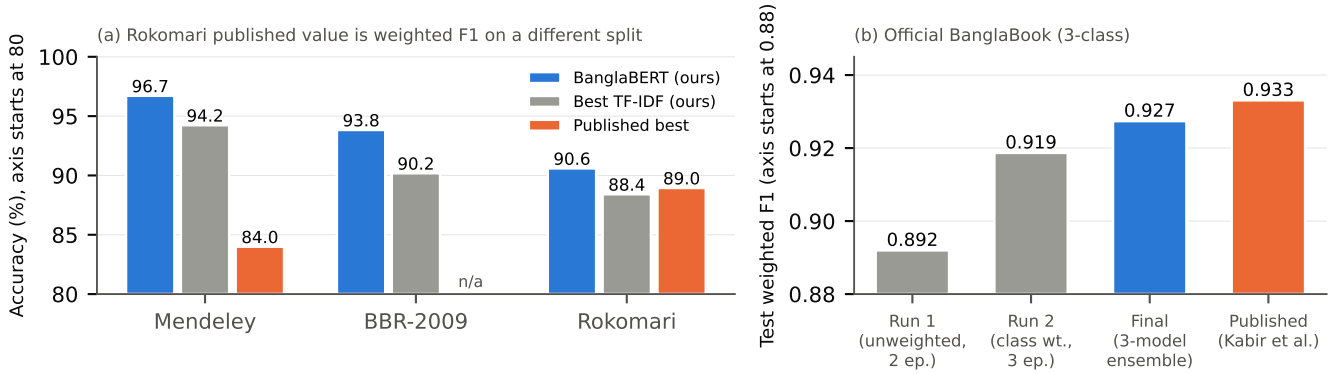


Fig. 3. (a) Our results against the best TF-IDF baseline and published results (the Rokomari published value is weighted F1 on a different split). (b) Official BanglaBook benchmark: weighted F1 of earlier runs, our final validation-selected ensemble and the published value.

TABLE II
PUBLISHED RESULTS AGAINST OURS. "COMPARABLE" SAYS WHETHER THE TWO NUMBERS COME FROM THE SAME DATA AND SPLIT.

Dataset (metric)	Published	Ours	Comparable?
Mendeley (acc.)	84.0% (Naive Bayes [15], authors' 2,000-review corpus)	**96.7%** BanglaBERT; 94.2% Naive Bayes	Indicative only: their corpus has 2,000 reviews, only 1,443 are public; Naive Bayes reaches 94.2% on the public part
Rokomari (wF1)	88.95% (XLM-R [16])	**89.97** $\pm$ **0.66%** BanglaBERT (their protocol, 3 shuffles)	No: their split cannot be reproduced; difference is within noise
BanglaBook (wF1)	0.933 (BanglaBERT-large [17])	0.927 (validation-selected ensemble)	Yes, official split. We are 0.6 points **below**

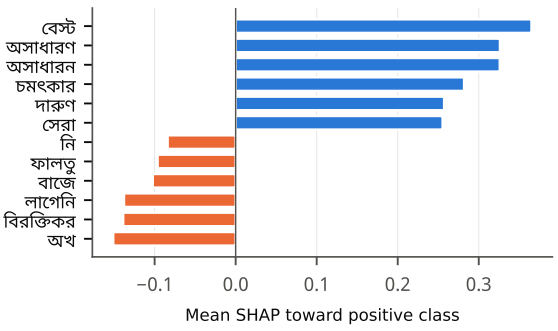


Fig. 4. Tokens with the largest mean SHAP contribution to the positive class (402 held-out reviews; orange pushes toward negative, blue toward positive).

Explanations can look convincing without being faithful, so we tested them. We deleted the k highest-SHAP tokens of each review and compared the drop in the predicted-class probability with deleting k random tokens (Fig. 5(a)). SHAP-selected tokens are three to five times more influential ($k = 1, 3, 5$: 0.120, 0.324, 0.444 against 0.024, 0.095, 0.150). Errors are not driven by negation words (4.7% with a negation word, 5.9% without) or by length (4.2 to 7.0% across quartiles). They are tied to the source instead: the error rate is 12.1% for BanglaBook reviews and 2.8% for Mendeley reviews.

E. Label noise and transfer

The source effect raises a question: are some of the "errors" actually wrong labels? Pooling out-of-fold predictions over three seeds and both model sizes, the ensemble misclassifies 111 of 2,009 reviews, and 69 of those errors (62%) were made with a probability above 0.9. A model that is that sure and still disagrees with the label deserves a second look. These cases are concentrated in the rating-derived source: 51 of 595 BanglaBook reviews (8.6%) against 18 of 1,414 Mendeley reviews (1.3%) (Fig. 5(b)).

To look closer, we shuffled the 69 suspected reviews together with 131 random ones, removed every label, and asked a single large language model to label them from the text alone. It judged 28 reviews unclear, which we dropped. On the random reviews it agreed with the dataset label 96.6% of the time (Cohen's $\kappa = 0.93$), so it is not simply contradicting the data. On the suspected reviews agreement fell to 25.9% ($\kappa = 0.00$), and 39 of the 43 reviews labeled negative were judged positive (Fig. 5(c)). Table III shows what the cases look like: three of the five are plainly positive reviews with a negative label, which is what one expects when a one- or two-star rating is turned into a label, while the other two are genuine model mistakes. So the list is a mix, not a clean set of label errors, and since the annotator is one LLM and not a human study, we treat this as a strong hint and not as ground truth.

Cross-source transfer tells a similar story (Fig. 6). A model trained on Mendeley reaches 84.5% on Rokomari, and one

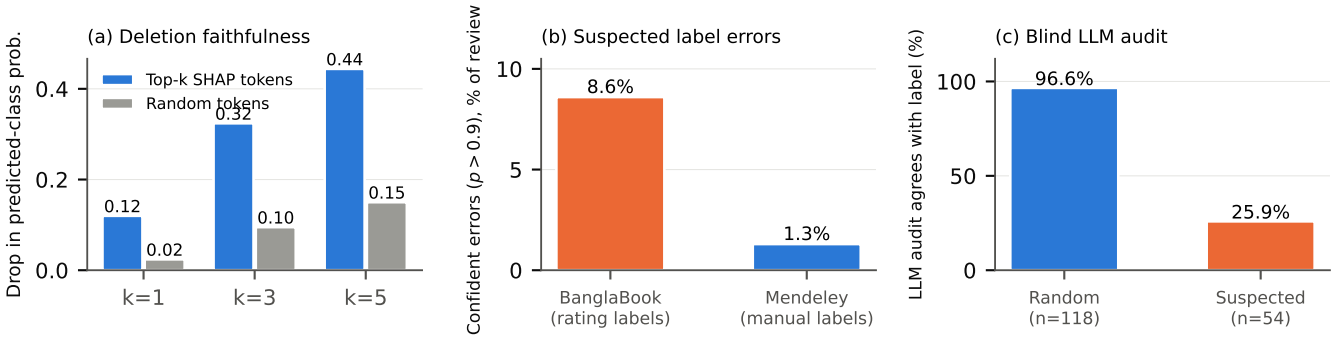


Fig. 5. (a) Deleting top-k SHAP tokens lowers the predicted-class probability much more than deleting random tokens. (b) Share of reviews on which the ensemble is confidently wrong ($p > 0.9$), by source. (c) Agreement between a blind LLM annotator and the dataset label for random and suspected reviews.

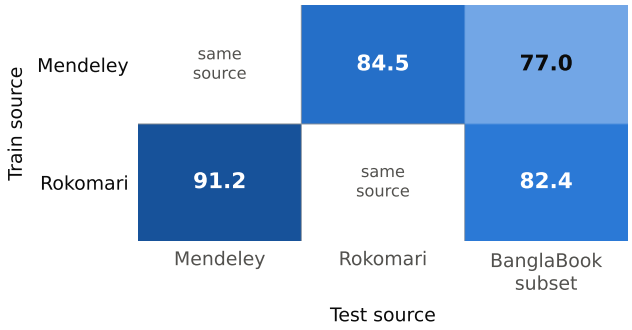


Fig. 6. Cross-source transfer of BanglaBERT (accuracy %, single seed). Models trained on a manually labeled or Rokomari source transfer poorly to the rating-labeled BanglaBook reviews.

trained on Rokomari reaches 91.2% on Mendeley, but both transfer poorly to the BanglaBook reviews (77.0% and 82.4% accuracy; macro-F1 58.3 and 63.8).

V. Discussion and Limitations

Three things stand out. First, a single split can mislead: one model scores anywhere from 73% to 86% depending on the fold. Second, it is pretraining on Bengali text, first of the embeddings and then of the whole encoder, and not recurrence, attention or model size, that separates the models, and a TF-IDF baseline is often stronger than the one used for comparison in earlier papers. Third, some of what looks like model error is label error, especially in rating-derived data, so accuracy on such corpora understates what a model can do.

BBR-2009 is small, rebuilt from two sources of different origin, and partly confounded by source. Rating-derived labels are noisy. The audit used one LLM annotator, so the suspected errors are not confirmed, and independent human annotation is the obvious next step. The Rokomari experiments used a different split from the published one, and on the official BanglaBook benchmark we remain 0.6 points behind. The recurrent and classical baselines were not tuned; pretrained embeddings narrow the gap to BanglaBERT but do not close it. Our Wilcoxon tests treat cross-validation runs as independent, which makes them somewhat optimistic. A language model audits the labels, and it may share blind spots with BanglaBERT. XLM-R and mBERT were tuned over a small grid only, and we did not test large language models with few-shot prompting.

VI. Conclusion

Under repeated cross-validation, recurrent models trained from scratch reach about 81% on Bengali book reviews, while fine-tuned BanglaBERT reaches 93.8% on our corpus and 96.7% on the Mendeley corpus. The explanations we checked are faithful, and many of the model's confident errors look like label errors. We hope the released splits and the honest comparison make future results easier to trust. Next we plan independent human re-annotation, few-shot large language models, and larger, manually verified Bengali review corpora.

Availability

The BBR-2009 text, labels, source flags, fold assignments and datasheet will be released for non-commercial use (CC BY-NC-SA 4.0, inherited from BanglaBook), together with code and logs, at https://github.com/NoorAbdullah02/bbr2009-bengali-book-review-sentiment.

Acknowledgment

We thank the creators of the Mendeley, BanglaBook and Rokomari review datasets for making their data available.

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TABLE III

EXAMPLES OF CONFIDENT DISAGREEMENTS (ENGLISH TRANSLATIONS OF THE BENGALI REVIEWS). THE FIRST THREE LOOK LIKE WRONG LABELS, THE LAST TWO LIKE MODEL MISTAKES.

Review (translated)	Dataset label	Model P(pos.)	Source	Likely cause
"Truly a very good book" (two heart-eyes emoji)	neg.	0.99	BanglaBook	label (rating)
"I liked it a lot, thank you to the author."	neg.	0.99	BanglaBook	label (rating)
"An extraordinary book; without it I would not have learned about many forbidden matters."	neg.	0.99	BanglaBook	label (rating)
"There is no equal to this book."	pos.	0.002	Mendeley	model: misread idiom with a negation word
"A good book, but I saw quite a few printing mistakes, and an alphabetical index would have helped."	pos.	0.004	Mendeley	model: mixed review

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